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Wave–Wind Misalignment and Drag Coefficient Rolloff in Tropical Cyclones

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1. Summary

We investigate wave-mediated air–sea exchange processes in a TC using the SCOAR fully coupled model simulations with modified COARE bulk flux parameterizations. Targeting Typhoon Khanun (2023), three different C_D rolloff formulations are considered evaluated against best-track intensity and drifting Spotter buoy wave observations.

The analysis focuses on two mechanisms:
(1) drag rolloff associated with MSS saturation
(2) directional modulation of surface stress by wind–wave misalignment.

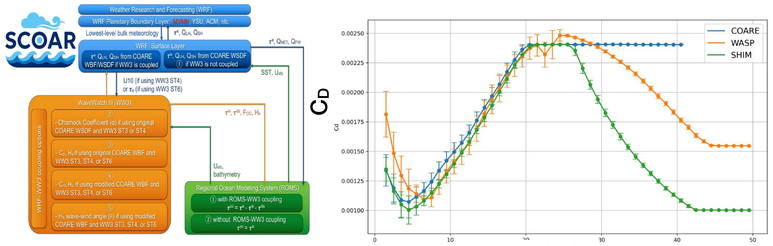
The main results are

1. Among the parameterizations tested, the observationally constrained formulation of Shimura et al. (2024) produces the best agreement with observed storm intensity and wave characteristics, with pronounced quadrant-dependent reductions in C_D and MSS.
2. The simulations capture an observed secondary relationship between wind–wave misalignment and MSS for wind speeds of 20–45 m/s, suggesting that cross-swell conditions reduce the efficiency of air–sea momentum transfer under extreme winds.
3. Three competing TC feedbacks are identified:
   (1) reduced C_D accelerates near-surface winds, favoring storm intensification;
   (2) reduced momentum transfer weakens upper-ocean cooling by suppressing turbulent mixing and TKE production, further supporting intensification; and
   (3) reduced enthalpy exchange coefficient (C_K) decreases together with C_D, reducing upward enthalpy fluxes despite stronger winds, favoring a weaker storm.

→ Overall, (1) & (2) dominate, yielding stronger storms characterized by higher maximum wind speeds and lower minimum sea-level pressure.

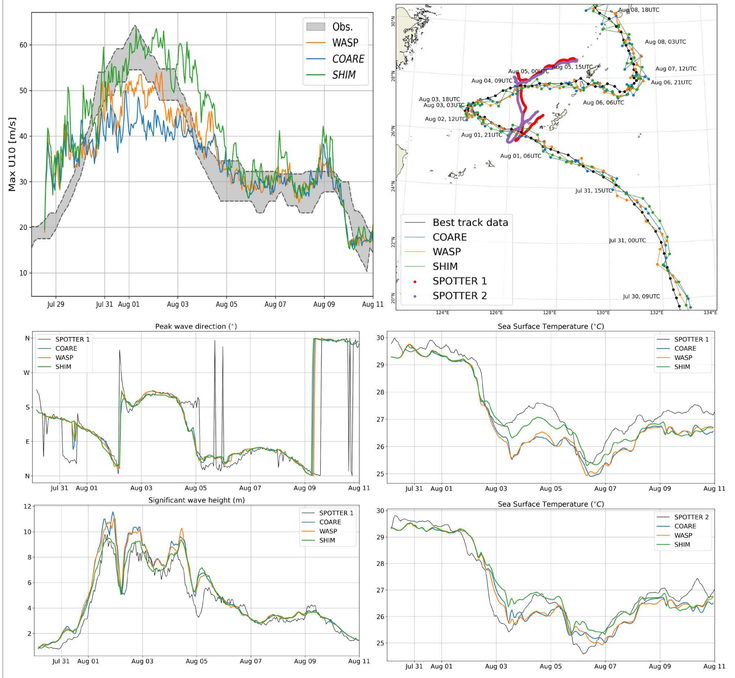
2. Model and Experiments

2.5km WRF-ROMS-WW3 simulations for Typhoon Khanun (2023) with 3 different drag coefficient rolloff cases

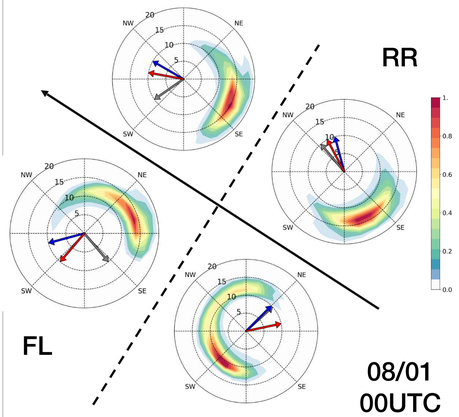


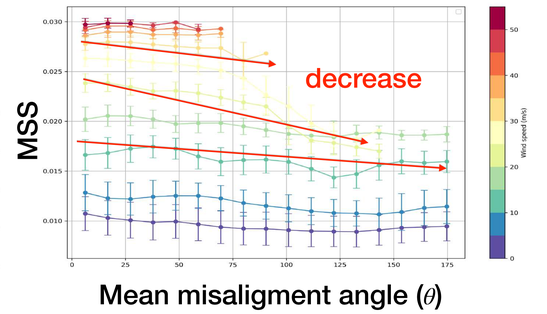
- COARE3.5 (Edson et al. 2013; Sauvage et al. 2023): $z_{rough} = H_s D\left(u_*/c_m\right)^B$ until cap (C_D~2.4 10^{-3})
- WASP (Bouin et al. 2024): $\alpha = A\chi^B$ until ~20 m/s, then decreases following observations
- SHIM (Shimura et al. 2024): COARE3.6 up to ~20 m/s; and cap u*~1.75 m/s for U_{10}>25 m/s

3. Typhoon Khanun (2023): Validation & Sensitivity



4. Wave-wind misalignment and mean-square slope



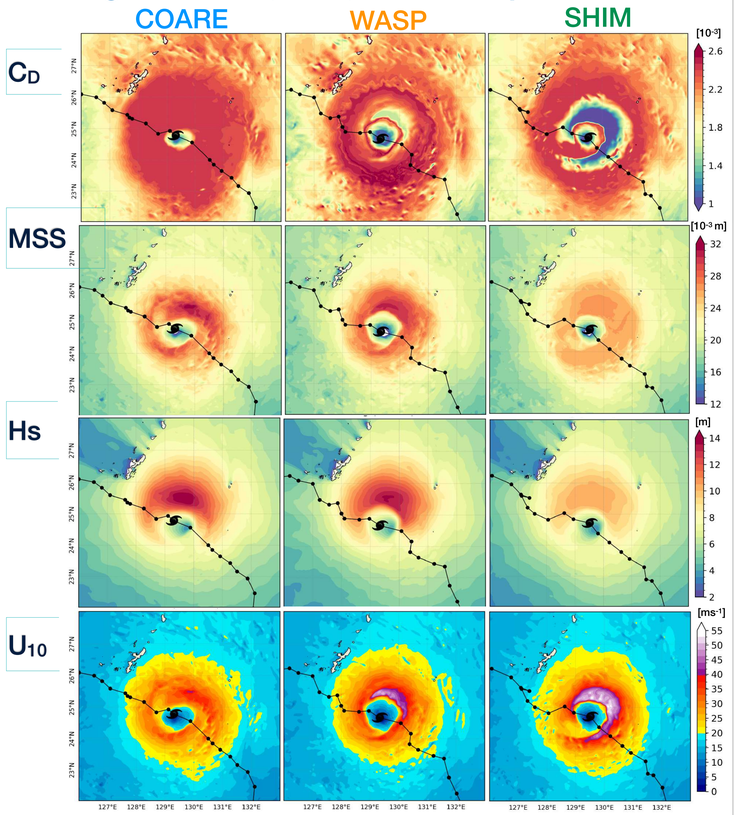


- Storm-quadrant dependence:
  - RR: Aligned swell, reduced directional spreading
  - FL: Cross-swell, large directional spreading

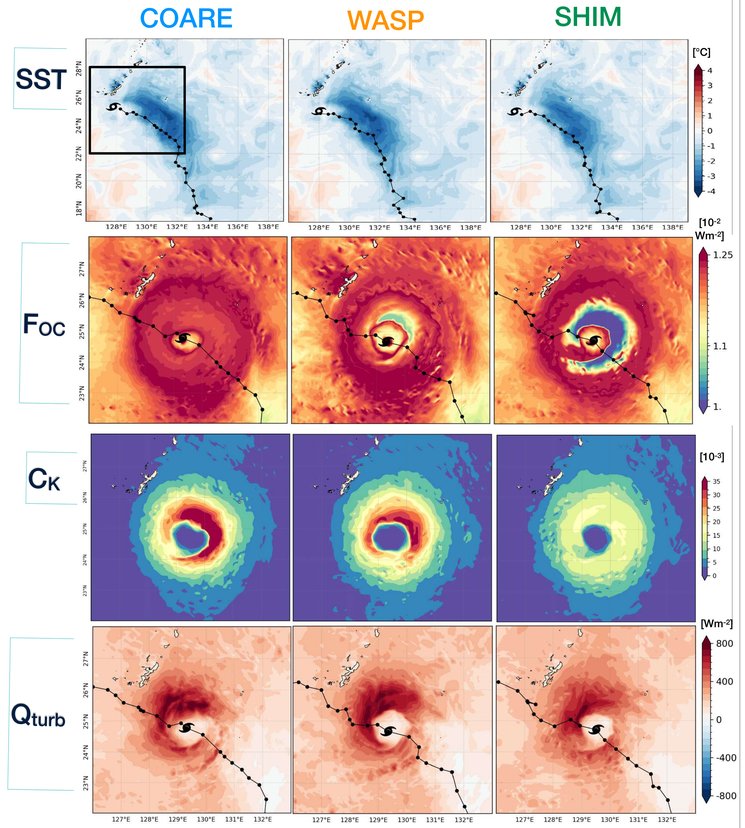
- Increasing MSS with increasing wind speeds
- Decreasing MSS with increasing wave-wind misalignment

→ Consistent with Davis et al. (2025), showing the secondary dependence of MSS on wind-wave misalignment

5. Drag coefficient, MSS, and wind speeds



6. SST and heat flux



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